

Optimizing energy consumption in centralized and distributed cloud architectures with a comparative study to increase stability and efficiency

Tangsen Huang^{*}, Weiguo Huang, Bin Zhang, Wenzhi Chen, Xuewen Pan

School of Information Engineering, Hunan University of Science and Engineering, Yongzhou 425199 Hunan, China

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ABSTRACT

Cloud computing serves as the backbone of modern digital infrastructure, supporting everything from basic data storage and processing to real-time big data analytics. However, the expanding scale and complexity of cloud systems have significantly increased energy consumption, posing serious environmental and economic concerns. This study explores an extended energy model that transitions from fully centralized to fully distributed architectures across diverse cloud-related configurations. The experimental results reveal that fully centralized architectures exhibit the highest energy consumption, with static energy usage from physical machines and switches accounting for approximately 1.4 MWh. In contrast, fully decentralized architectures demonstrated energy efficiency improvements, reducing energy consumption by 19% to 28%. An important observation was that static energy consumption remained consistent across all architectures, while dynamic energy consumption varied depending on traffic loads and system configurations. In fully distributed systems with central controllers, the dynamic energy consumption of physical machines remained below 0.1 MWh, and switches and routers consumed less than 0.05 MWh. These findings highlight the potential of decentralized architectures in achieving substantial energy savings, especially in edge computing and Internet of Things (IoT) applications. By reducing energy consumption, these architectures contribute to both environmental sustainability and cost efficiency. This study offers valuable insights for cloud service providers aiming to optimize energy efficiency across different architectural models.

1. Introduction

The modern computing landscape faces a vital challenge of increasing energy consumption, with cloud computing solutions being confronted with it. Data centers are key energy consumers in cloud computing infrastructure. A report by the Lawrence Berkeley National Laboratory in 2016 showed that data centers consume around 70,000 MWh, equivalent to about 1.8 % of the total electricity consumption in the United States. [1] Predictions indicate further increase in the future, with an estimated increase to about 73,000 MWh by 2030 in the United States.

Simultaneously, the new wave of Internet of Things applications requires more distributed cloud architectures. These architectures, often referred to as fog, edge, and sometimes peer-to-peer infrastructures, are designed to meet the strict requirements of low latency and high bandwidth anticipated by Internet of Things applications [2,3]. While the inevitability of deploying such infrastructures is undeniable, the energy implications of these architectures remain largely unknown.

Cloud computing has evolved towards more distributed architectures due to increasing demand for mobility, powerful data processing capabilities, and reduced latency. A significant milestone in this path occurred in 2014 when the European Telecommunications Standards Institute introduced the concept of mobile edge computing [4]. Mobile edge computing, initially designed to provide IT and cloud computing capabilities in radio access networks, quickly evolved. By 2016, the Mobile Edge Computing Group at the European Telecommunications Standards Institute expanded its scope and renamed itself as Multi-access Edge Computing intending to provide heterogeneous network support [5]. Managing mobile edge computing infrastructures varies depending on specific use cases, ranging from hierarchical control structures to distributed control structures [6,7]. It is worth noting that mobile edge computing environments may combine mobile equipment to provide computational services to smaller devices when necessary [7].

In a parallel development, the increase in Cisco's Internet of Things system led to the proposal of the fog computing concept in 2012 [8]. Fog computing, defined as a horizontal architecture at the system level by

^{*} Corresponding author.

E-mail address: huangtangsen@huse.edu.cn (T. Huang).

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Nomenclature		VM	Virtual Machine
<i>Abbreviation</i>		VMC	Virtual Machine Consolidation
AR	Autoregressive	<i>Symbols</i>	
ARIMA	Auto-Regressive Integrated Moving Average	E	Electricity consumption
BAS	Building Automation System	P	Power Consumption
CEWOAVM	Cost-efficient Whale Optimization Algorithm for Virtual Machine	T	Time
CFL	Centralized Federated Learning	$L.NP_T$	Total number of processed bytes
CHP	Combined Heat and Power	$DC_{(i)}$	Total traffic received by data center
CIFAR	Canadian Institute for Advanced Research	U^H	Constant value
CPU	Central Processing Unit	U^{CN}	The average CPU usage
CVRMSE	Coefficient of Variance of Root Mean Squared Error	U^{VM}	The average CPU usage on each virtual machine
DFL	Decentralized Federated Learning	N_{Core}	Core routers
ETS	Exponential Smoothing	<i>Subscripts</i>	
FL	Federated Learning	α	Individual physical machines
HDD	Hard Drive Disk	β	Individual network switches
LSTM	Long-term Short-Term Memory	γ	Individual network elements
MBE	Mean Bias Error	<i>Superscripts</i>	
MNIST	Modified National Institute of Standards and Technology	H	The hypervisor
ROGI	Goal of Optimizing Latency and Power Consumption	CN	Computational Node
SLA	Service Level Agreement	VM	Virtual Machine
SOFC	Solid Oxide Fuel Cell	<i>Greek Symbols</i>	
SSD	Solid State Drive	μ	Service rate
TM	Thermal Management	λ	Packet arrival rate
UACO	Using the Enhanced Ant Colony Optimization		

the OpenFog Consortium, coordinates computing, storage, control, and network resources across the chain from the cloud to IoT devices. Its main goal is to manage time-sensitive and large-volume data generated at the network edge by IoT devices [9]. With the expansion of cloud access to IoT endpoint devices and their provisioning with computational, storage, and networking capabilities, fog computing effectively bridges the gap between cloud and things [8]. Structurally, the fog computing architecture adopts a hierarchical approach divided into three layers: IoT devices, fog nodes, and cloud data centers [9]. Managing fog nodes based on program needs with options such as centralized or distributed models is different. Amid this complexity, efforts like the Open Edge Computing Glossary initiated by Linux, with evolving precise definitions in this landscape, aim to provide clarity [10].

Recent research has shed light on the outlook of cloud simulators and data centers, showing over 20 simulators in the past decade that CloudSim [11] has emerged as a prominent competitor. Despite its advantages, CloudSim, with inherent limitations, especially in communication and network energy modeling, struggles with modeling large-scale data centers, federated structures, and placing multiple virtual machines.

1.1. Literature review

Yao et al. [12] conducted a study on daylighting and energy efficiency of DS-STPV windows in the cold regions of China. By simulation and empirical data, they analyzed the design parameters such as photovoltaic cell coverage, orientation of the window, and window-to-wall ratio. They concluded that a 30 % photovoltaic cell coverage, south orientations, and 30 % window-to-wall ratio are the optimal conditions that can be met for both natural lighting provision and energy savings. This paper described the huge potentiality of the DS-STPV to improve building energy efficiency and reduce interior glare while providing better indoor environmental quality and guidelines for practical green building design in cold climates. Hosamo et al. [13] addressed issues related to optimizing energy efficiency and the occupants' comfort in buildings by considering 12,000 energy simulations

for Tvedestrand Upper Secondary School in Norway. The performance comparison showed that the Multi-Layer Perceptron outperformed other machine learning models. The sensitivity analyses showed that ventilation rate, room depth, façade U-value, and heat gains have the most influencing factors on energy consumption. The results are compared to traditional HVAC systems considering Building Management System solutions; the latter has shown operational cost reductions when the investment costs were 1200 NOK/m² and annual costs were 150 NOK/m². Optimized parameters achieved energy consumption ranging between 11.05 and 22.51 kWh/m² with zero discomfort hours, hence giving valuable guidance toward sustainable and economically feasible building design. Pan et al. [14] have proposed a new algorithm that fuses Lyapunov-based control with LSTM networks and PSO on a layered fog computing network, completely opening a new way of achieving optimal latency and resource optimization in IoT systems. It forecasts users' task features, including priorities of unloading tasks, and then dynamically adjusts resources in rechargeable networks to further save energy. Simulation results demonstrate great improvements compared with traditional schemes in both latency reduction and optimization of resource utilization. That would be an effective way to improve the efficiency of IoT systems and users' experiences facing increasing demands for interconnected smart services. Shi et al. [15] developed a building energy-saving renovation optimization method by automated machine learning and the NSGA-III algorithm. The method achieved an accuracy of more than 95 % in prediction, where Stacked Ensemble and Gradient Boosting Machine performed better among all algorithms used. Climate change affected the efficiency of the renovation measures with energy savings between 22.06 % and 26.48 % and improvements in thermal comfort from 31.87 % to 43.73 % under current scenarios; these gains were even better under future conditions. Such a method could balance effectively the contradiction among energy consumption, cost, and comfort, indicating its potential contribution to promoting building renovation energy savings, especially in hospitals. As and Bilir [16] evaluated energy consumption in hospitals in Turkey, focusing on Aksaray and Bursa provinces, using 1,440 scenarios generated from the library of the Revit BIM software. Using machine learning algorithms,

especially artificial neural networks, energy consumption, CO₂ emissions, costs, and life cycle expenditures were effectively estimated. R-squared (R^2) values were between 0.95 for energy, 0.94–0.97 for CO₂ emissions, and 0.93–0.97 for costs. Hence, the study showed very good prediction accuracy. Consequently, in this study, the model was able to simplify the process of energy evaluation according to TS 825-compliant thermal transmittance value and building orientation angle, which can be used all over Turkey to optimize hospital energy consumption. Katal et al. [17] addressed the issue of energy-aware container placement in cloud data centers by using the modified Discrete Firefly Algorithm and Discrete Firefly with Local Search Mechanism. The results demonstrated that, compared with classic algorithms, DFFLSM can save up to 40.85 % of the average energy consumption and up to 21.89 % active physical machines in the heterogeneous environment. This paper justifies that DFFLSM is effective in improving resource utilization while minimizing service-level agreement violations within containerized cloud computing systems.

Table 1 summarizes all the reviewed literature. In this table, an attempt has been made to present the main focus of the literature, their key findings and the challenges they face.

1.2. Research gaps

Given the context of this paper and its increasing importance, it would be useful to examine a number of research gaps in this field. This paper also attempts to address all of these gaps as much as possible.

The first research gap that can be mentioned is the energy inefficiency in centralized cloud architectures. Comparing different architectural models and demonstrating the energy efficiency of distributed and semi-distributed systems is of great importance due to the high energy consumption in cloud architectures, which is the focus of this work. The lack of integrated energy models is another research gap. Previous studies lack a comprehensive model for integrating static and dynamic energy consumption. This study attempts to introduce a hybrid energy model for the overall assessment of energy consumption in different architectural structures. The lack of empirical validation is another gap that is evident in previous studies. The next gap in this field is the limited insight into the impact of traffic on energy consumption. Understanding the impact of traffic patterns and network configuration on energy dynamics by analyzing static and dynamic consumption under different loads and network configurations is of great importance, which has been less addressed in previous studies. Insufficient energy solutions for IoT and edge computing can lead to high latency and low energy efficiency. The research gaps in this field are not limited to these issues, but due to the nature of any scientific work that has a number of limitations, this paper also considers these issues more and tries to answer them.

1.3. Innovations

Cloud-related architectures represent a critical area of research, with the need for innovative solutions becoming increasingly pressing as the demand for cloud computing services grows. The core contribution of this study lies in the development of a novel classification framework and energy consumption model, providing a much-needed tool for evaluating the energy efficiency of various cloud architectures from fully centralized systems to fully decentralized ones. This innovation extends beyond theoretical contributions, offering practical insights that cloud service providers can use to improve sustainability in their operations. The present paper presents a holistic energy model for analysis and optimization of both static and dynamic energy consumptions of different cloud architectures: fully centralized, semi distributed, fully distributed, and distributed with central controllers. By combining virtualization, workload distribution, and hybrid static-dynamic modeling, this research achieved scalable and accurate estimates for energy consumption given different traffic and network configurations. Empirical

Table 1

The summary of the reviewed literature.

Ref.	Focus Area	Key Insights	Challenges Highlighted
[18]	Energy-efficient resource allocation in cloud computing	Energy-aware algorithms enhance SLA performance while optimizing resource use.	Difficulty in achieving energy efficiency without SLA violations.
[19]	Forecasting building energy consumption	Incorporating operational variables in LSTM models improves prediction accuracy.	Managing diverse input data for optimal model performance.
[20]	Energy optimization and load balancing in cloud computing	CEWOAVM algorithm reduces energy consumption significantly while improving VM migration efficiency.	Balancing optimization of VM migration with energy efficiency.
[21]	Balancing electricity production and consumption	Categorizing forecasts by operational periods improves accuracy in energy modeling.	Adapting methods to specific operational timeframes.
[22]	Green technologies for cloud data centers	Virtualization and system-level management reduce environmental impact.	Sustaining efficiency while addressing long-term ecological effects.
[23]	Resource management in cloud IoT systems	Frameworks for workload modeling and energy management improve IoT application performance.	Managing resource complexity in large-scale IoT-cloud environments.
[24]	Latency and power optimization in fog computing	ROGI model partially offloads computation to optimize energy use.	Managing energy efficiency across various network layers.
[25]	Thermal-aware resource allocation in data centers	TM-VMC prevents hotspots while reducing SLA violations.	Ensuring real-time performance under thermal constraints.
[26]	Distributed control in IoT-enabled BAS	Robust control ensures performance even under communication uncertainties.	Handling network failures in distributed IoT systems.
[27]	Energy prediction for Docker containers	ETS and ARIMA models improve accuracy for specific container configurations.	Enhancing prediction for varied container setups.
[28]	Energy usage in data centers	Comparative analysis of cloud and on-premises setups reveals energy sustainability gaps.	Improving efficiency without compromising scalability.
[29]	Decentralized Federated Learning (FL)	Fedstellar platform reduces training time and improves efficiency in decentralized setups.	Mitigating communication bottlenecks in FL systems.
[30]	Renewable energy in microgrids	SOFC integration in hybrid systems achieves high energy efficiency.	Addressing integration challenges with existing grid systems.
[31]	Database technologies in e-commerce	Blockchain ensures security, while scalable storage addresses performance demands.	Balancing security, scalability, and computational efficiency.
[32]	Runtime load balancing in cloud systems	Efficient algorithms enhance dynamic load redistribution and resource utilization.	Maintaining system stability during real-time optimization.

validation over the French Grid 5000 testing platform demonstrated that such improvements really achieve a gain of between 19 and 28 % in energy savings with respect to fully decentralized architectures compared to a centralized one. In particular, this decentralized architecture turns out to be very effective for enhancing energy efficiency in dynamic components, such as physical machines and switches, thus becoming very suitable in IoT/Edge computing where low latency and energy efficiency are of paramount importance. The study also set out ways through which energy-efficient architectural strategies could be adopted by cloud service providers to balance operational costs

efficiently with sustainability concerns. Showing how decentralized systems could avoid energy wastage, thereby efficiently managing the resources dynamically, the research work pointed out the importance of the trend for future cloud infrastructures. These recommendations are based on real-time AI-enabled energy optimization, hybrid architecture, and integration of renewable sources of energy. This research work strongly motivates the design of cloud systems that are both ecologically sustainable and cost-effective, hence providing practical insights into ways of improving energy efficiency in meeting demands imposed on digital infrastructures.

1.4. Paper organization

The continuation of the sections of the article is as follows. The second part includes structure modeling and related methodologies. In the third part, the study case and the obtained results are presented. In the fourth part, a brief but complete discussion is given. In the final section, general conclusions are presented.

2. Modeling

2.1. Static consumption model

Cloud data centers, essential to modern digital infrastructure, incur significant residual energy costs even during idle periods. Minimizing this static energy consumption is crucial for achieving sustainability and cost efficiency [27]. The primary contributors to static energy usage are physical machines and network devices. Physical machines consume energy continuously to remain operational, even without processing tasks, while network devices such as switches and routers exhibit baseline energy consumption, known as standby power, driven by components like power supplies and cooling systems. Static energy consumption can be quantified by multiplying the power usage of each device by the operational time interval. For data centers, Eq. (1) defines the total static energy consumption over time T , encompassing physical machines and switches. Similarly, Eq. (2) calculates the static consumption of telecommunication networks, considering all routers connecting data centers and end users. This model enables a comprehensive analysis of static energy usage, paving the way for optimizing data center architectures for reduced environmental and operational costs:

$$E_{DC(i)}^{\text{Static}} = T \cdot \left[\sum_{\alpha \in CN(i)} P_{(\alpha)}^{\text{IDLE}} + \sum_{\beta \in SN(i)} P_{(\beta)}^{\text{IDLE}} + \sum_{\gamma \in NE(i)} P_{(\gamma)}^{\text{IDLE}} \right] \quad (1)$$

$$E_{\text{Net}}^{\text{Static}} = \sum_{r \in RT} T \cdot P_{(\beta)}^{\text{IDLE}} \quad (2)$$

For scenarios that use energy management techniques, some devices may go into sleep mode instead of remaining idle. In such cases, Eq. (1) and Eq. (2) should be adjusted in a way that reflects lower energy consumption related to sleep mode for those specific devices. This parameter intentionally removed from the equations for clarity. Eq. (3) combines the energy consumption of the telecommunications network while stationary with the total static energy consumption of each data center, and combines it with the cooling energy consumption shown by the electricity consumption efficiency metric. This equation provides a comprehensive picture of the overall stationary energy consumption for the entire architecture of the cloud data center. By analyzing and optimizing these components of stationary energy consumption, cloud service providers can significantly reduce their environmental impacts and operating costs. The following section examines the dynamic energy consumption of cloud data centers resulting from processing workloads and data traffic.

$$E_{\text{Total}}^{\text{Static}} = E_{\text{Net}}^{\text{Static}} + \sum_i \left[PUE_{(i)} \cdot E_{DC(i)}^{\text{Static}} \right] \quad (3)$$

2.2. Dynamic energy consumption of the data center

The fat-tree network topology architecture is selected for the proposed system due to its scalability, fault tolerance, and efficiency. This topology provides a hierarchical structure that enables optimal utilization of network resources and minimizes congestion. The fat-tree design allows for efficient routing of traffic, which is crucial for maintaining high system performance under heavy load conditions. Additionally, its inherent scalability ensures that the system can adapt to the growing demands of the network. The fault-tolerant nature of the fat-tree topology ensures continuous operation even in the event of node or link failures, further enhancing the system's reliability. Given these factors, the fat-tree topology is deemed an ideal choice for the system's goals of scalability, efficiency, and robustness in diverse network environments. Dynamic energy consumption in cloud data centers arises primarily from internal network operations, particularly in centralized architectures. Key contributors include [28]:

- **Packet Processing:** Energy required for tasks like header parsing and routing decisions.
- **Load Movement:** Energy used to store and transfer data through the network's internal infrastructure.

A two-parameter model, incorporating energy consumption per packet ($E_{\text{Switch}}^{\text{PKT}}$) and per byte ($E_{\text{Switch}}^{\text{SF}}$), accurately captures this behavior. In a fat-tree network topology, where identical switches simplify analysis, the total number of bytes processed over a time interval is expressed as $L \cdot NP_T$ representing the number of processed packets. The dynamic energy consumption of a single switch is computed as shown in Eq. (4), providing a framework for understanding and optimizing network-related energy costs in data centers.

$$E_{\text{Switch}}^{\text{Dynamic}} = NP_T \cdot E_{\text{Switch}}^{\text{PKT}} + L \cdot NP_T \cdot E_{\text{Switch}}^{\text{SF}} \quad (4)$$

To express the dynamic energy consumption of all switches in a data center in the time period T , we can use Eq. (5). This equation takes into account the total number of switches in the data center. Ultimately, the total energy consumed by the physical machines' dynamic energy consumption can be calculated according to Eq. (6).

$$E_{DC(i)}^{\text{Dynamic}} = |NE_{(i)}| \cdot E_{\text{Switch}}^{\text{Dynamic}} \quad (5)$$

$$E_{DC(i)}^{\text{Dynamic}} = \sum_{\alpha \in CN(i)} E_{(\alpha)}^{\text{Dynamic}} + \sum_{\beta \in SN(i)} E_{(\beta)}^{\text{Dynamic}} \quad (6)$$

To estimate network traffic controlled by a switch, we can use the assumption of uniform distribution of virtual machines among physical machines. This means that the traffic generated by each specific physical machine is distributed equally among the data center for other physical machines. Based on this assumption, Eq. (7) helps us calculate P , which represents the probability of traffic originating from a physical machine located in the same data center for other physical machines.

$$P = \frac{PM_{(i)} - 1}{\sum_i [PM_{(i)} - 1]} \quad (7)$$

Traffic generated by end users also contributes to the overall network load. We denote μ as the average traffic rate generated by each user directed towards data centers. With this value, we can calculate the total traffic received by data center $DC(i)$ from end users over time T as $(U \cdot T \cdot \mu) \times (8 \cdot L \cdot N)^{-1}$, where U represents the total number of end users and N represents the total number of data centers.

The overall processed traffic in a data center includes five key components:

- 1) Traffic is sent outbound to end users.

- 2) Internal traffic exchanged between resident physical machines.
- 3) Traffic sent to other data centers.
- 4) Traffic received from other data centers.
- 5) Traffic from end users.

Taking these components into account, Eq. (8), NP_T , expresses the average number of packets processed by a switch over time T . It calculates this value by dividing the total traffic managed by all switches in $DC_{(i)}$ during time T by the number of switches and averages it. Ultimately, by combining Eq. (6) (which represents the dynamic energy consumption of physical machines) and Eq. (5) (which represents the dynamic energy consumption of all switches), we arrive at Eq. (15). This equation calculates the total dynamic energy consumption of data centers. This comprehensive model enables data center operators to understand and optimize the energy requirements of their internal networks and pave the way for a more sustainable and cost-effective cloud infrastructure.

$$NP_T = \frac{X_1 + X_2 + X_3 + X_4 + X_5}{X_6} \quad (8)$$

$$X_1 = 3 \cdot \lambda \cdot Q \cdot PM_{(i)} \quad (9)$$

$$X_2 = 5 \cdot \lambda \cdot P \cdot (1 - Q) \cdot PM_{(i)} \quad (10)$$

$$X_3 = 3 \cdot \lambda \cdot (1 - P) \cdot (1 - Q) \cdot PM_{(i)} \quad (11)$$

$$X_4 = 3 \cdot (N - 1)^{-1} \cdot \sum_{j \in \{1, \dots, N\} \setminus \{i\}} [\lambda \cdot (1 - P) \cdot (1 - Q) \cdot PM_{(j)}] \quad (12)$$

$$X_5 = 3 \cdot N^{-1} \cdot U \cdot \mu \quad (13)$$

$$X_6 = 10 \cdot L \cdot n^2 \quad (14)$$

$$E_{DC_{(i)}}^{\text{Dynamic}} = PUE_{(i)} \cdot [E_{DC_{(i)}-NE}^{\text{Dynamic}} + E_{DC_{(i)}-PM}^{\text{Dynamic}}] \quad (15)$$

2.3. Dynamic energy consumption of physical machines

Dynamic energy consumption in cloud data centers fluctuates with server workloads, primarily driven by CPU utilization. Research identifies CPU activity as the main contributor to this energy variation. Using multicore server data, the relationship between CPU usage and dynamic energy consumption is modeled with high accuracy, often within a few percent for tested applications [29–31]. This model estimates dynamic energy by measuring incremental energy increases as server cores transition from idle to full capacity. Such insights enable optimized infrastructure design, improving responsiveness and cost efficiency in cloud environments. For a physical machine (denoted by x) equipped with k cores, we can record the average energy consumption during scenarios with fully loaded cores. Subtracting the idle power consumption of the physical machine $P_{(x)}^{\text{IDLE}}$ from these measurements separates the dynamic component. These dynamic energy consumption values are defined as $P_{(x,j)}^{\text{Cores}}$, where j varies from 0 to k . The proposed model assumes a relationship between each of these measured power consumption values. Therefore, for a physical machine utilizing a fraction of its total CPU capacity, if the Eq. (16) holds, where j varies from 0 to $k-1$, then the dynamic energy consumption of physical machines x during the time period T can be expressed as shown in Eq. (17) [33].

$$\frac{j}{K} \leq U \leq \frac{j+1}{K} \quad (16)$$

$$E_{(x)}^{\text{Dynamic}} = T \cdot \left[\left((j+1) \cdot P_{(x,j)}^{\text{Cores}} - j \cdot P_{(x,j+1)}^{\text{Cores}} \right) + K \cdot U \cdot \left(P_{(x,j+1)}^{\text{Cores}} - P_{(x,j)}^{\text{Cores}} \right) \right] \quad (17)$$

To estimate the average CPU usage on each computational node, we can combine the average CPU usage on each virtual machine with the

number of virtual machines available on each computational node. In addition, a constant value U^H is included to calculate the energy consumption of the hypervisor and other processes related to the cloud that monitor physical machines. This combined approach guides us to Eq. (18) [33].

$$U^{CN} = U^H + \frac{V \cdot U^{VM}}{\sum_i |CN_{(i)}|} \quad (18)$$

Ultimately, by using Eq. (17) and considering all physical machines inside data centers, we can express the total dynamic energy consumption related to physical machines in that data center using Eq. (6). By analyzing and optimizing these factors affecting dynamic energy consumption, cloud service providers can effectively scale their infrastructures to meet workload requirements while minimizing energy waste. Different cloud system architectures exhibit varying levels of energy efficiency due to differences in resource allocation, hardware requirements, and system configurations. Fully centralized architectures tend to consume more energy because they rely on a larger number of physical machines, switches, and routers. This results in higher static energy consumption, particularly from the physical infrastructure, which remains constant regardless of system load. In contrast, decentralized architectures demonstrate greater energy efficiency due to their distributed nature, which allows for more dynamic resource allocation based on demand. This adaptability reduces idle energy consumption and ensures that resources are utilized more efficiently. Furthermore, decentralized systems enable better load balancing and more efficient traffic management, which further decreases energy consumption. For example, in fully distributed systems with central controllers, the dynamic energy consumption of physical machines is significantly lower (below 0.1 MWh), while energy usage for switches and routers is even lower (under 0.05 MWh). This flexibility in energy consumption patterns makes decentralized architectures more energy-efficient, especially in scenarios like edge computing and Internet of Things (IoT) applications, where demand is highly variable and distributed processing helps reduce energy waste.

2.4. Dynamic energy consumption of the telecommunication network

Cloud architectures interact with the energy needs of the telecommunication network linking data centers and users, making energy optimization crucial for sustainable, cost-effective infrastructure. Like data center switches, network routers consume dynamic energy based on data volume. This energy is driven by two factors: 1) packet processing, where routers analyze and route incoming data packets, and 2) load movement, where additional energy is consumed for temporary storage and transfer of data within the network. The energy consumption of a router depends on its type due to the difference in hardware capabilities and processing power. To calculate the dynamic energy consumption of a telecommunication network in time interval T , Eq. (19) can be used. Here, $NP_{T(r)}$ represents the average number of packets processed by router r in time interval T , $E_{(r)}^{\text{SF}}$ represents the energy required to process a packet by router r , and $E_{(r)}^{\text{PKT}}$ represents the energy for storing and transmitting a byte by router r [34].

$$E_{\text{Net}}^{\text{Dynamic}} = \sum_{r \in \text{RT}} \left[L \cdot NP_{T(r)} \cdot E_{(r)}^{\text{SF}} + NP_{T(r)} \cdot E_{(r)}^{\text{PKT}} \right] \quad (19)$$

For a more detailed analysis, we can classify routers into four types: Core, Backbone, Metro, and Feeder. The total number of routers is equal to the sum of these separate categories. By decomposing Eq. (19) for each group of routers, we arrive at Eq. (20) [34].

$$E_{\text{Net}}^{\text{Dynamic}} = E_{\text{Core}}^{\text{Dynamic}} + E_{\text{Backbone}}^{\text{Dynamic}} + E_{\text{Metro}}^{\text{Dynamic}} + E_{\text{Feeder}}^{\text{Dynamic}} \quad (20)$$

$$E_{\text{Backbone}}^{\text{Dynamic}} = N_{\text{Backbone}} \cdot [L \cdot NP_{T-\text{Backbone}}(r) \cdot E_{\text{Backbone}}^{\text{SF}}(r) + NP_{T-\text{Backbone}}(r) \cdot E_{\text{Backbone}}^{\text{PKT}}(r)] \quad (21)$$

$$E_{\text{Core}}^{\text{Dynamic}} = N_{\text{Core}} \cdot [L \cdot NP_{T-\text{Core}}(r) \cdot E_{\text{Core}}^{\text{SF}}(r) + NP_{T-\text{Core}}(r) \cdot E_{\text{Core}}^{\text{PKT}}(r)] \quad (22)$$

$$E_{\text{Metro}}^{\text{Dynamic}} = N_{\text{Metro}} \cdot [L \cdot NP_{T-\text{Metro}}(r) \cdot E_{\text{Metro}}^{\text{SF}}(r) + NP_{T-\text{Metro}}(r) \cdot E_{\text{Metro}}^{\text{PKT}}(r)] \quad (23)$$

$$E_{\text{Feeder}}^{\text{Dynamic}} = N_{\text{Feeder}} \cdot [L \cdot NP_{T-\text{Feeder}}(r) \cdot E_{\text{Feeder}}^{\text{SF}}(r) + NP_{T-\text{Feeder}}(r) \cdot E_{\text{Feeder}}^{\text{PKT}}(r)] \quad (24)$$

In order to estimate the average number of packets processed by each router, we need to analyze the traffic routes and the corresponding probabilities. There are three main types of traffic that pass through the telecommunications network:

- I. Traffic between data centers: Data exchange is done directly between data centers.
- II. User traffic to data center: Data flowing from end users to data centers.
- III. Data center-to-user traffic: Data that returns from data centers to end users.

Eq. (25) to Eq. (27) show this general traffic flow [35].

$$\left(NP_T^{\text{DC-DC}} = \frac{T}{8 \cdot L} \sum_i [\lambda \cdot (1-p) \cdot (1-q) \cdot PM_{(i)}] \right) \quad (25)$$

$$\left(NP_T^{\text{EU-DC}} = \frac{U \cdot T \cdot \mu}{8 \cdot L} \right) \quad (26)$$

$$\left(NP_T^{\text{DC-EU}} = \frac{T}{8 \cdot L} \cdot \frac{\sum_i [\lambda \cdot q \cdot PM_{(i)}]}{\frac{T}{8 \cdot L}} \right) \quad (27)$$

Different cloud architectures have different traffic patterns. Hence, we can consider three common scenarios [35]:

- 1) Fully centralized architecture: In this architecture, a data center is connected to a single-core router ($p = 0$). There are two potential traffic paths between the data center and end users. Considering the N_{Core} core routers and users equally distributed among the feeders, the first path is selected in one of the N_{Core} cases and the second path in N_{Core}^1 of the N_{Core} cases. Therefore, Eq. (28) to Eq. (30) are applied for the fully centralized architecture.

$$NP_{T-\text{Core}}^{\text{FC}}(r) = [NP_T^{\text{EU-DC}} + NP_T^{\text{DC-EU}}] \cdot \left[\frac{1}{N_{\text{Core}}} + 2 \cdot \frac{N_{\text{Core}}-1}{N_{\text{Core}}} \cdot N_{\text{Core}} \right] \quad (28)$$

$$NP_{T-\text{Backbone}}^{\text{FC}}(r) = \frac{[NP_T^{\text{EU-DC}} + NP_T^{\text{DC-EU}}]}{N_{\text{Backbone}}} \quad (29)$$

$$NP_{T-\text{Feeder}}^{\text{FC}}(r) = \frac{[NP_T^{\text{EU-DC}} + NP_T^{\text{DC-EU}}]}{N_{\text{Feeder}}} \quad (30)$$

- 2) Semi-distributed architecture: Table 2 helps us calculate the number of processed packets by each router for the semi-distributed architecture, which the corresponding equation is shown in Eq. (31) to Eq. (34).

Table 2

Probabilities of paths in semi-distributed architecture.

DC-DC paths	Probability	User-DC paths	Probability
BCCB	$\frac{3}{2N_{\text{Core}}}$	FMBCCB	$\frac{3}{N_{\text{Core}}}$
BCB	$\frac{1}{2N_{\text{Core}}} - \frac{1}{2N_{\text{Metro}}}$	FMBCB	$\frac{1}{N_{\text{Core}}} - \frac{1}{N_{\text{Metro}}}$
BB	$\frac{1}{2N_{\text{Metro}}}$	FMB	$\frac{1}{N_{\text{Metro}}}$
MBCCB	$\frac{3}{N_{\text{Core}}}$	FMBCCBM	$\frac{3}{N_{\text{Core}}}$
MBCB	$\frac{1}{N_{\text{Core}}} - \frac{1}{N_{\text{Metro}}}$	FMBCBM	$\frac{1}{N_{\text{Core}}} - \frac{1}{N_{\text{Metro}}}$
MB	$\frac{1}{N_{\text{Metro}}}$	FM	$\frac{1}{N_{\text{Metro}}}$
MBCCBM	$\frac{3}{2N_{\text{Core}}}$		
MBCBM	$\frac{1}{2N_{\text{Core}}} - \frac{1}{2N_{\text{Metro}}}$		
MBM	$\frac{1}{2N_{\text{Metro}}}$		

F: Feeder router, M: Metro router, B: Backbone router, C: Core router, DC: Data Center.

$$NP_{T-\text{Core}}^{\text{PD}}(r) = NP_T^{\text{DC-DC}} \cdot \left[\frac{14}{N_{\text{Core}}} - \frac{2}{N_{\text{Metro}}} \right] + [NP_T^{\text{EU-DC}} + NP_T^{\text{DC-EU}}] \cdot \left[\frac{14}{N_{\text{Core}}} - \frac{2}{N_{\text{Metro}}} \right] \quad (31)$$

$$NP_{T-\text{Backbone}}^{\text{PD}}(r) = NP_T^{\text{DC-DC}} \cdot \left[\frac{14}{N_{\text{Core}}} - \frac{3}{N_{\text{Metro}}} \right] + [NP_T^{\text{EU-DC}} + NP_T^{\text{DC-EU}}] \cdot \left[\frac{14}{N_{\text{Core}}} - \frac{3}{N_{\text{Metro}}} \right] \quad (32)$$

$$NP_{T-\text{Metro}}^{\text{PD}}(r) = NP_T^{\text{DC-DC}} \cdot \left[\frac{e}{N_{\text{Core}}} \right] + [NP_T^{\text{EU-DC}} + NP_T^{\text{DC-EU}}] \cdot \left[1 + \frac{4}{N_{\text{Core}}} - \frac{1}{N_{\text{Metro}}} \right] \quad (33)$$

$$NP_{T-\text{Feeder}}^{\text{PD}}(r) = \frac{[NP_T^{\text{EU-DC}} + NP_T^{\text{DC-EU}}]}{N_{\text{Feeder}}} \quad (34)$$

- 3) Fully distributed architecture: Similarly, Table 3 presents the number of packets for the fully distributed architecture, which leads to Eq. (35) to Eq. (38).

$$NP_{T-\text{Core}}^{\text{FD}}(r) = NP_T^{\text{DC-DC}} \cdot \left[\frac{14}{N_{\text{Core}}} - \frac{2N_{\text{Metro}}}{N_{\text{Feeder}}} \right] + [NP_T^{\text{EU-DC}} + NP_T^{\text{DC-EU}}] \cdot \left[\frac{14}{N_{\text{Core}}} - \frac{2N_{\text{Metro}}}{N_{\text{Feeder}}} \right] \quad (35)$$

Table 3

Possibilities of routes in completely distributed architecture.

DC-DC paths	Probability	User-DC paths	Probability
FMBCCBMF	$\frac{6}{N_{\text{Core}}}$	FMBCCBMF	$\frac{6}{N_{\text{Core}}}$
FMBCBMF	$\frac{2}{N_{\text{Core}}} - \frac{2N_{\text{Metro}}}{N_{\text{Feeder}}}$	FMBCBMF	$\frac{2}{N_{\text{Core}}} - \frac{2N_{\text{Metro}}}{N_{\text{Feeder}}}$
FMF	$\frac{2N_{\text{Metro}}}{N_{\text{Feeder}}} - \frac{1}{N_{\text{Feeder}}}$	FMF	$\frac{2N_{\text{Metro}}}{N_{\text{Feeder}}} - \frac{1}{N_{\text{Feeder}}}$
F	$\frac{1}{N_{\text{Feeder}}}$	F	$\frac{1}{N_{\text{Feeder}}}$

$$NP_{T-Backbone}^{FD}(r) = NP_T^{DC-DC} \cdot \left[\frac{16}{N_{Core}} - \frac{2N_{Metro-1}}{N_{Feeder}} \right] + [NP_T^{EU-DC} + NP_T^{DC-EU}] \cdot \left[\frac{16}{N_{Core}} - \frac{2N_{Metro-1}}{N_{Feeder}} \right] \cdot \frac{1}{N_{Metro}} \quad (36)$$

$$NP_{T-Metro}^{FD}(r) = NP_T^{DC-DC} \cdot \left[\frac{16}{N_{Core}} - \frac{4N_{Metro-1}}{N_{Feeder}} \right] + [NP_T^{EU-DC} + NP_T^{DC-EU}] \cdot \left[\frac{16}{N_{Core}} - \frac{2N_{Metro-1}}{N_{Feeder}} \right] \cdot \frac{1}{N_{Metro}} \quad (37)$$

$$NP_{T-Feeder}^{FD}(r) = NP_T^{DC-DC} \cdot \left[\frac{16}{N_{Core}} - \frac{1}{N_{Feeder}} \right] + [NP_T^{EU-DC} + NP_T^{DC-EU}] \cdot \left[\frac{16}{N_{Core}} - \frac{1}{N_{Feeder}} \right] \cdot \frac{1}{N_{Feeder}} \quad (38)$$

By integrating Eq. (15) representing the dynamic energy consumption of physical machines and Eq. (40) representing the dynamic energy consumption of a telecommunication network, we obtain the total dynamic energy consumption of a cloud-related architecture. This comprehensive model empowers cloud service providers to understand and optimize the energy needs of their network infrastructure, making the cloud environment more sustainable and cost-effective. The total energy consumption for each architecture is the sum of its static and dynamic components, as presented in Eq. (3) and Eq. (39) respectively. This comprehensive approach facilitates accurate estimation of energy consumption in different network architectures and increases efficiency and sustainability in cloud infrastructure management [36].

$$E_{Feeder}^{Dynamic} = \sum_i [E_{Net}^{Dynamic} + E_{DC(i)}^{Dynamic}] \quad (39)$$

2.5. Hybrid static-dynamic energy consumption model

While static and dynamic energy consumption are often treated separately, integrating both factors provides a more accurate and holistic view of a data center's energy behavior. One way to combine both static and dynamic energy is to use a hybrid model that accounts for their simultaneous contributions. This combined model can be represented as:

$$E_{Total}^{Hybrid,T} = E_{DC}^{Static,T} + E_{DC}^{Dynamic,T} \quad (40)$$

Since virtualization is widely used in cloud environments to improve resource utilization, incorporating the impact of virtualization efficiency into energy models can offer more refined energy consumption estimates. The virtualization efficiency factor (η) could be defined as the ratio of virtual machine (VM) workload to the host machine's capacity. The energy consumed by a physical machine (PM) with virtualization can be adjusted as:

$$E_{PM}^{Dynamic} = E_{PM}^{Dynamic} \times \eta \quad (41)$$

$$\eta = \frac{U_{VM}}{U_{PM}} \quad (42)$$

Cooling energy consumption is an important component of total energy usage in data centers. As the workload increases, cooling systems need to be more active to maintain an optimal temperature. A simple model to account for this adjustment based on load is:

$$E_{Cooling}(T) = E_{Cooling}^{Base} \times \left(1 + \beta \times \frac{L}{L_{max}} \right) \quad (43)$$

where, $E_{Cooling}^{Base}$ is the base cooling energy consumption (when the data center is idle or lightly loaded). L is the current workload or traffic volume. L_{max} is the maximum possible workload or traffic that the system can handle. β is a constant representing the sensitivity of cooling energy consumption to load.

Energy costs can vary depending on the time of day and the energy provider. To account for these fluctuations, a time-varying energy cost model can be introduced. The total energy cost $C(T)$ for a given period T is expressed as:

$$C(T) = \int_T E_{Total}^{Hybrid}(T) \times p(t) dt \quad (44)$$

3. Case studies and presentation of results

To facilitate a fair and comprehensive comparison, we create a baseline model that is evaluated in all cloud architectures. This baseline model includes 2000 computational nodes that work for one hour. It is important to acknowledge that architectures such as fully distributed and semi-distributed with central controllers require more powerful computational nodes due to support for storage nodes and additional services.

- Fully centralized: This architecture consolidates all 2000 data centers into one large-scale data center.
- Scattered half: This configuration uses 8 data centers with average and uniform size, each containing 250 data centers.
- Completely distributed with central controllers: This architecture uses 2000 nano data centers, each equipped with a central controller. In addition, 8 central controller data centers manage small storage nodes.
- Completely distributed: This configuration also utilizes 2000 nano data centers, each of which houses a data center within itself.

Table 3 has summarized these architectural configurations for easy reference. The number of physical machines based on storage node architecture varies depending on the architecture and the number of data centers it supports. For fully centralized and semi-distributed architectures, we assume 3 global storage nodes for high availability, along with additional storage nodes for every 2000 data centers. In a fully distributed architecture with centralized controllers, each of the 8 data centers has a central controller, placing 3 central nodes within itself. The fully distributed architecture uses a single-purpose and multi-purpose physical machine in each nano data center.

To ensure accurate energy consumption specifications, we conducted tests using the French Grid 5000 testing platform. In these experiments, Taurus servers with 12 cores, 32 GB memory, and 598 GB storage were used for physical machines in fully centralized and semi-centralized architectures. For fully distributed and fully distributed architectures with central controllers, Parasilo servers with 16 cores, 128 GB memory, and a combination of Hard Drive Disk (HDD) and SSD memories were used as physical machines, with power consumption values loaded on each core. The Grid 5000 testing system measures power consumption through external watt meters on each server and provides a value per second with high precision. Fig. 1 shows the results of accurate measurements. The energy consumption for both servers increases progressively as the number of active cores increases from 0 to 16. While for Taurus servers, the energy consumption increased linearly up to about 230 MWh with 16 active cores, Parasilo servers kept the energy consumption similar to Taurus only up to about 11 cores. Values above this showed Parasilo started consuming more energy, well above the consumption by Taurus when at about 240 MWh for 16 active cores. Both

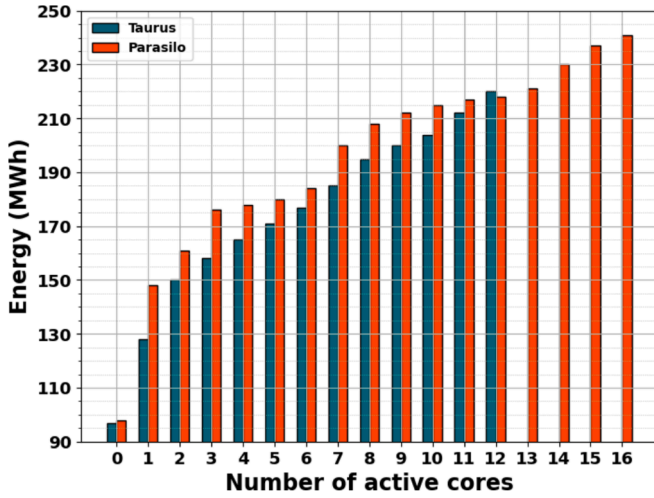


Fig. 1. Energy profiles of physical machines.

seem to have similar efficiencies up to 11 active cores, after which ParaSilo becomes inefficient since at higher core counts disproportionate amounts of energy are consumed wrt Taurus. Most astoundingly, both machines seem to consume around 90 MWh when zero cores are active; perhaps this accounts for the background energy consumption just to keep the machines on. This perhaps suggests that for higher counts of active cores, Taurus servers are much more power-effective than 11, which could be indicative of a number of factors in relation to the architecture: optimizations on Taurus for centralized as opposed to Parasilo for distributed.

Network topology in each data center is based on different architectures. A fully centralized architecture with its own large data center uses a 32-year-old Fat tree switch topology with 640 switches. A semi-distributed architecture with fewer data centers uses a 16-year-old Fat tree topology. A fully distributed architecture with central controllers only requires one switch in each data center for central control. Interestingly, a fully centralized architecture eliminates the need for a network within the data center.

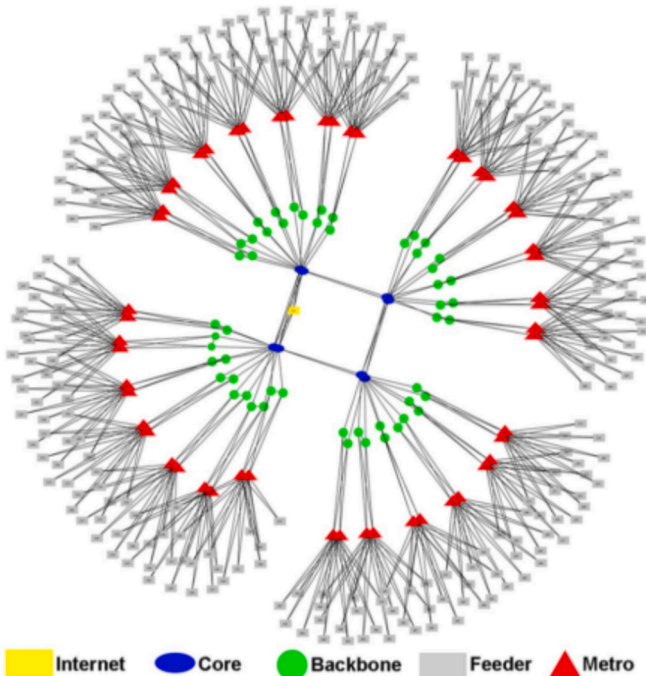


Fig. 2. The structure of the system's telecommunication network.

Inspired by the simplified ISP network presented in Fig. 2, we consider a more realistic scenario of a telecommunications network in this work, describing data transfers between data centers and users. It consists of 16 Core routers, 104 Backbone routers, 104 Metro aggregation routers, and 520 of Feeder routers, summed up to 744 routers interconnected by 1436 links. This would be the detailed network design to establish a level playing field of comparison and representation of energy consumption and performance exhibited with various cloud architectures. With this set of baseline inter-relationships in consideration, a consistent network configuration would really enable us to investigate the inherent characteristics related to energy efficiency and performance effectively with every cloud architecture, thus enabling informed choices of cloud solutions as required.

For scenario-based virtual machines with small and homogeneous virtual machines, we assume that each virtual machine is allocated a CPU core reflecting typical settings in cloud deployment. In addition, we adopt an average CPU utilization rate of 15 % for these virtual machines. This value is based on real-world measurements conducted by a large cloud provider [37]. Given the similarity in processing power between the cores in Taurus and Parasilo servers, we assume uniform core performance in both types of servers.

Estimating the average CPU utilization for storage nodes presents a greater challenge due to the variety of workloads and the number of active storage nodes. To maintain simulation simplicity, we allocate 50 % average CPU utilization to storage node-based physical machines. For router energy consumption, we use detailed measurements from reference [38]. These measurements show static energy consumption of 12000 W for Core routers, 4500 W for Backbone routers, and 2100 W for Metro and Feeder routers.

For data center switches, we simulate a 16-port switch model based on Fat tree topology. Specifically, is considered a Cisco WS-C6503 switch equipped with a 16-port line card WS-X6516-GE-TX. According to Cisco's power calculator, this configuration consumes approximately 280 W. Referring to [39], we assume that 70 % of total traffic in both mega and micro data centers remains within the data center itself.

Based on the experimental studies presented in [40], we observe at least minor changes in energy consumption in each packet in different routers. Therefore, we create a uniform value of 1350 nJ/pkt for all routers and switches in our simulation. Similarly, we adopt a constant value of energy consumption in each stored and transmitted byte, 14 nJ/Byte, for all network elements. For energy efficiency, we consider a value of 1.2 for all mega and micro data centers in fully centralized and semi-distributed architectures. This value is consistent with industry standards, such as the average energy efficiency reported by Google of 1.11 for their data centers. To isolate the impact of architectural changes on energy consumption, we disregard the effects of various energy management techniques such as sleep control and device active states in our simulations. Under this assumption, all physical machines and network elements remain active throughout the simulation period. These resource consumption profiles will enable us to devise a realistic simulation environment for conducting experiments in studying energy efficiency and performance characteristics of different cloud architectures. Thus, service providers will be in a better position through such profiling to make informed choices for the selection of an architectural approach that best fits their needs.

This section focuses on energy consumption differentiation in four cloud architectures: fully centralized, semi-distributed, fully distributed with central controllers, and fully distributed under varying traffic loads. We analyze both static and dynamic energy consumption components to gain a deeper understanding of factors affecting energy efficiency. Fig. 3 highlights the static energy consumption characteristics for each architecture. Static energy consumption for computational nodes in fully centralized and fully distributed architectures with central controllers is partially higher due to the use of a variety of physical machine sizes with different energy consumption levels.

Other major contributors to the static energy consumption are the

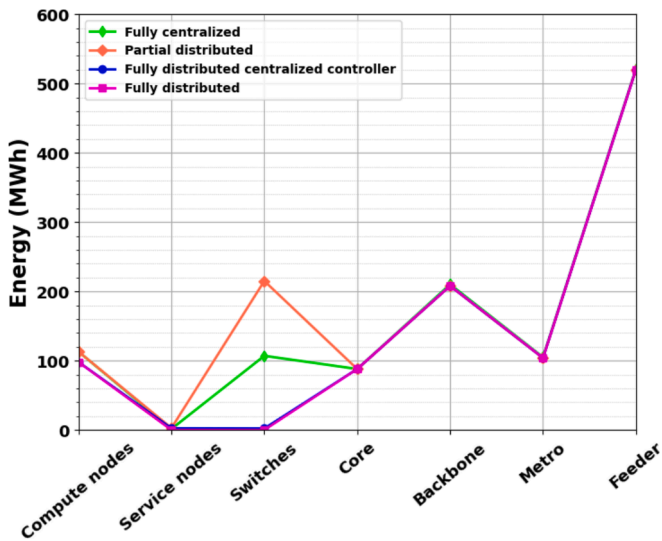


Fig. 3. Static energy consumption.

storage nodes. Semi-distributed and fully distributed architectures with central controllers show higher static energy consumption due to the greater number of storage nodes used compared to completely centralized architectures. It is worth mentioning that a fully distributed architecture with central controllers, despite having more storage nodes, consumes less overall static energy for storage nodes, which contributes to improving the efficiency of electricity consumption. Semi-distributed architectures, due to their dependence on a larger number of switches, waste most of the static energy. On the other hand, a fully distributed architecture eliminates the need for an in-data center network, completely avoiding this issue. Telecommunication network routers maintain a constant level of static energy consumption in all architectures. Main routers, while individually consuming more energy, are outnumbered by Feeder routers, which dominate the total static energy consumption for this component. Several important points can be observed based on the results obtained. The number of in-data center switches is a critical factor in static energy consumption in cloud architectures. Other important factors include the number of storage nodes, energy efficiency of physical machines, and overall electricity consumption efficiency of data centers. Fig. 3 shows the static energy consumption across various network components compute nodes, service nodes, switches, core, backbone, metro, and feeder under four architectural setups: fully centralized, partial distributed, fully distributed with a centralized controller, and fully distributed. The fully centralized and the partially distributed topologies have quite low consumption in most of the equipment, peaking to around 200 MWh on the switches, while the fully distributed one presents much higher energy consumptions, peaking over 500 MWh on the feeder component. The fully distributed centralized controller architecture balances between the extremes, with energy levels close to the fully centralized architecture but spiking in the backbone. Compute and service nodes show minimal energy consumption in the fully centralized and partial distributed models, while switches demand the most energy in these setups. Fully distributed architecture has much higher energy consumption in backbone, metro, and core, out of which the feeder has a maximum. Overall, the fully centralized architecture is the most energy-efficient, while the fully distributed architecture is the least energy-efficient, especially in downstream components like the feeder.

Figs. 3-6 demonstrates the all-dynamic energy consumption for each architecture under different traffic loads and packet lengths. At low traffic rates, the dynamic energy is somewhat insensitive to the packet's length. Nevertheless, a noticeable impact is observed at higher traffic rates. A completely scattered architecture shows the lowest dynamic

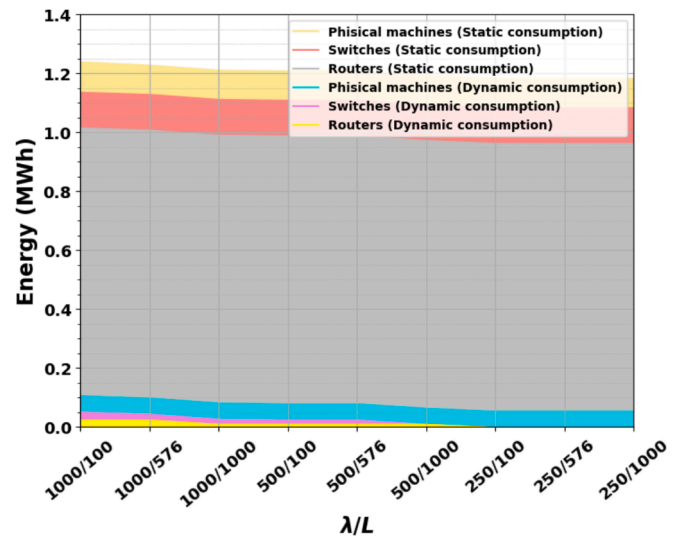


Fig. 4. Total energy consumption of the system in a fully centralized structure.

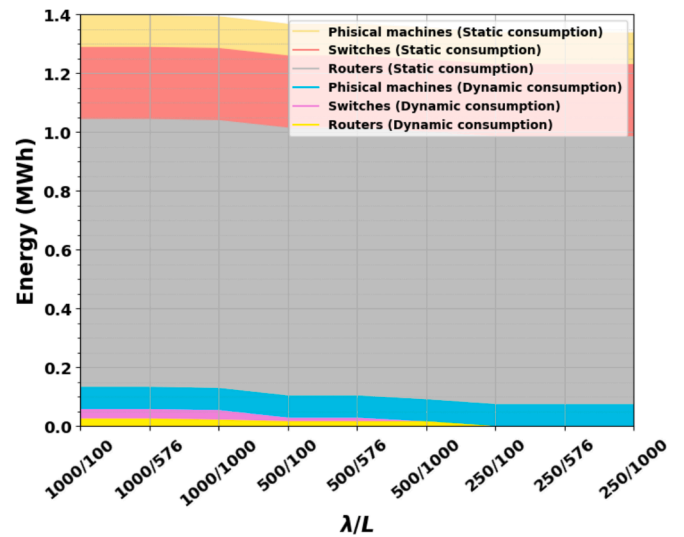


Fig. 5. Total energy consumption of the system in the semi-dispersed structure.

energy consumption, primarily due to the lack of a network within the data center. Although completely scattered architectures with central controllers and fully scattered use more energy storage nodes, resulting in higher dynamic energy consumption than physical machines, these architectures compensate in other areas. A completely scattered architecture with central controllers requires fewer switches to connect its storage nodes, leading to lower dynamic energy consumption compared to completely centralized and semi-scattered architectures. Interestingly, as the traffic mounts up, semi-scattered architecture is advantageous concerning dynamic energy consumption rather than a centralized architecture. This is because larger data centers in a completely centralized architecture require longer data paths between physical machines, leading to increased energy consumption.

As shown in Figs. 7-10, the dynamic energy consumption associated with routers is directly related to the locations of data centers and traffic patterns. Although a completely distributed architecture shows slightly lower dynamic energy consumption for routers compared to fully distributed with central controllers, a more noticeable difference can be expected for applications with heavy traffic flow between controllers and storage nodes.

Key insights from dynamic energy consumption analysis are as

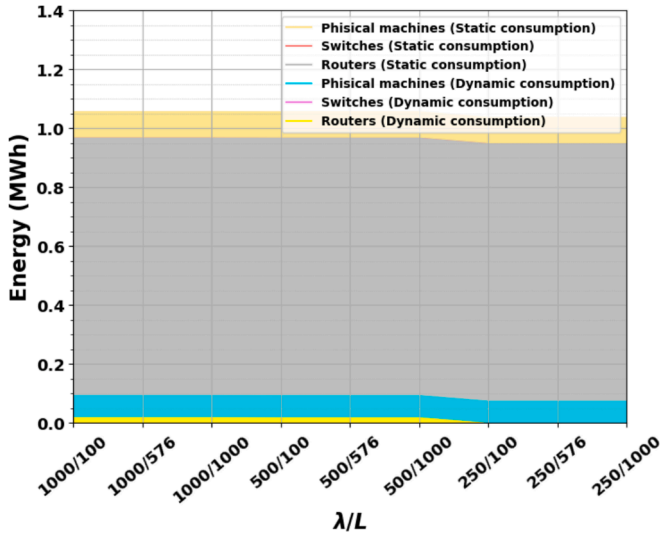


Fig. 6. Total system energy consumption in a completely distributed structure with central controllers.

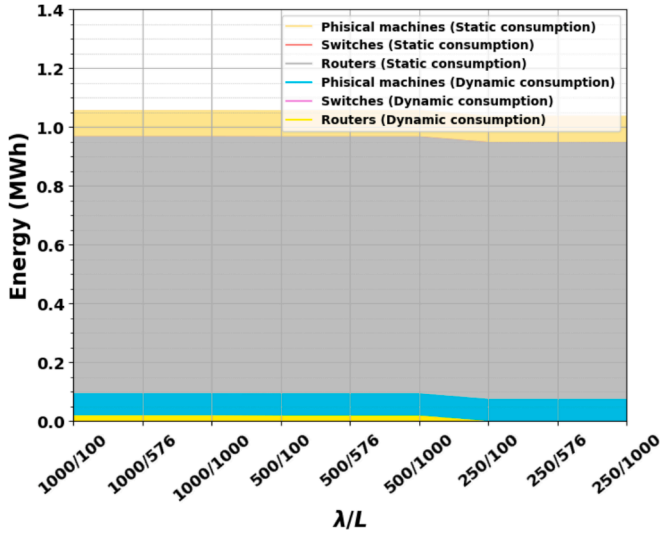


Fig. 7. Total energy consumption of the system in a completely distributed structure.

follows:

- The type and number of physical machines used in architecture are very important in determining dynamic energy consumption.
- Network topology and the number of network elements significantly affect dynamic energy consumption.
- The effectiveness of overall electricity consumption in data centers plays a vital role.
- Traffic patterns and data transfer volumes are essential factors that must be considered.

Fig. 4 shows the total energy consumption of a fully centralized system, breaking down static and dynamic energy consumption for physical machines, switches, and routers across varying values of λ/L . The largest part of the energy profile is static consumption, basically dominated by the physical machines, including switches and followed by routers. However, dynamic consumption is much lower and depends insignificantly on the number of physical machines, switches, and routers. As can be seen from the decline in λ/L , the total energy

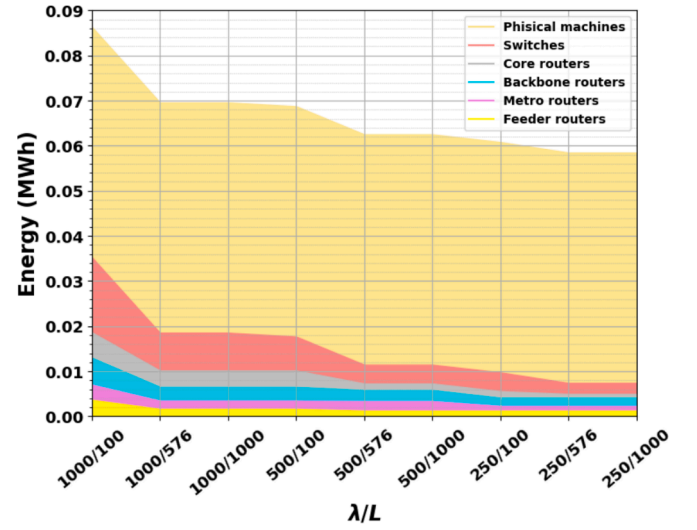


Fig. 8. Dynamic energy consumption of the system in a fully centralized structure.

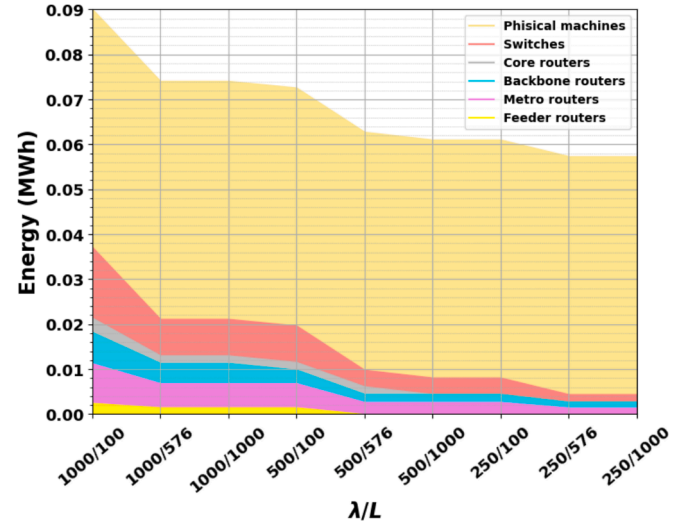


Fig. 9. Dynamic energy consumption of the system in the semi-dispersed structure.

consumption has remained somewhat constant though the static consumption does; therefore, the share of dynamic energy use is negligible against the static energy in fully centralized systems. On the whole, it is affected by static consumption. Hence, most of the energy consumed in the system is by physical machines and switches. Even with changes in λ/L , the energy profile remains largely unchanged, suggesting that centralized systems offer limited flexibility in reducing energy costs.

Fig. 5 illustrates the total energy consumption of the semi-dispersed system as an area chart stacked by energy contributions with respect to static and dynamic consumption of physical machines, switches, and routers. In all energy consumptions, the consumption of the static energy is dominant. Energy consumption of physical machines generated by the static energy consumption is about 1.0 to 1.2 MWh, which slightly decreases when the load ratio λ/L decreases. The estimated contributions of the switches and routers are in the range of 0.1 to 0.2 MWh each. For smaller loads, consumptions are a bit lower. Dynamic consumptions are far smaller. Physical machines contribute approximately 0.05 MWh dynamically, while switches and routers contribute approximately 0.01 MWh each. From the highest load case-1000/100, the overall energy consumption is about 1.3 to 1.4 MWh and decreases gradually to about

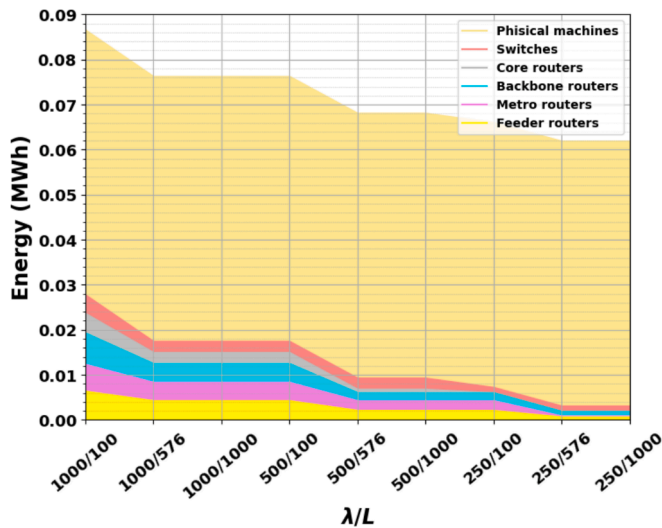


Fig. 10. Dynamic energy consumption of the system in a completely distributed structure with central controllers.

1.0 MWh at the lowest load, 250/1000. This diagram shows the very important position of the static energy consumption in the system's energy consumption.

Fig. 6 depicts a stacked area chart representing the entire energy usage (in MWh) of a system characterized by a fully distributed architecture with centralized controllers. Energy consumption is categorized into static and dynamic components for physical machines, switches, and routers. Routers account for the majority of energy use, consistently providing around 1.0 MWh across all λ/L ratios. While the virtual machines will consume dynamically, the physical machines stand at about 0.1 MWh, that of the switch is even lower, standing at 0.05 MWh. Regarding dynamic consumption, routers, switches, and physical devices contribute little quantities, each below 0.05 MWh. The overall energy consumption begins at around 1.1 MWh at the maximum load ratio (1000/100) and diminishes somewhat to 1.0 MWh at the minimum load ratio (250/1000). This research indicates that static energy consumption, especially from routers, constitutes the predominant factor in the system's total energy usage, whereas dynamic consumption is negligible.

Fig. 7 represents the overall energy consumption of the system in the fully distributed architecture, considering the static and dynamic energy consumptions of physical machines, switches, and routers. Along the x-axis, it plots a range of ratios representing λ over L , while on the y-axis, it plots energy consumption in MWh. Yellow, red, and blue lines represent static consumption, the value of which for all λ/L ratios was about the same: the biggest consumers of energy were physical machines, routers, and switches. Dynamic consumption, on the other hand, is rather low compared with static consumption, including very minor variations across the λ/L ratios represented by cyan, magenta, and yellow-green areas. The dynamic consumption of the physical machines is higher than switches and routers but very small with respect to the general static consumption. As the ratio λ/L decreases, there is a general mild decrease of the total energy consumption; hence, smaller values of λ/L would imply marginal improvements in the energy efficiency of this system. Generally, static energy consumption dominates in the energy profile of the system, while the dynamic one contributes minimally.

Fig. 8 showing the dynamic energy consumptions in a fully centralized system: the contribution made by physical machines and switches, core, backbone, metro, and feeder routers. The major contributors of dynamic energy are physical machines – the yellow area. While λ/L decrements, the ratio within physical machines tumbles. The main switches come into play again in the red area where energy consumption was gradually lowered. Core routers (cyan), backbone routers (blue),

metro routers (magenta), and feeder routers (green) are far less dominant in this figure and even here energy consumption lowers with greater centralization of the system—that is, with a lower value of λ/L . The graph shows that centralizing a system result in reduced dynamic energy consumption by all components while increasing its energy efficiency due to the low λ/L ratio, especially within the physical machines and switches.

Fig. 9 reflects the dynamic energy consumption from a system that has a semi-dispersed structure where the energy contribution among physical machines, switches, and all types of routers is considered. The x-axis represents various ratios of λ/L while on the y-axis, energy consumption was measured in MWh. The physical machines have the prime share in the dynamic energy consumptions while their consumption decreases as the ratio λ/L reduces. The contribution of switches within this particular region is very high; their energy uses go down with the increased dispersion of the system. For core, backbone, metro, and feeder routers, consumptions are much lower, still decreasing when the ratio λ/L goes down. This graph, in general, goes down to the consumption of dynamic energy throughout all components if the system goes more toward dispersion—meaning a big plus in energy efficiency, especially in physical machines and switches, which can be shown similarly in fully centralized systems.

Fig. 10 depicts the dynamic energy consumptions of a system in a completely distributed architecture using central controllers, by showing the contribution of physical machines, switches, and different types of routers—cores, backbone, metro, and feeder. The yellow color corresponds to physical machines, the dominant in dynamic energy consumptions, whose usage decreases as the ratio decreases. The switches contribute considerably, too, but energy consumption is observed to go down appreciably with higher distribution of the system. Core routers—colored cyan, backbone routers—blue, metro routers—magenta, and feeder routers green consume considerably lesser power as compared to physical machines and switches, but these, too, are observed to go down progressively with the decrease in the λ/L ratio. Overall, dynamic energy consumed decreases through all components as the level of distribution of the system increases, indicating a higher energy efficiency for more distributed systems using central controllers, which confirms several trends seen in the analysis of other structures.

Fig. 11 also presents the dynamic power consumed by the system inside a fully disaggregated architecture showing shares of physical machines, switches, and various routers: core, backbone, metro, and feeder. Yellow area: physical machines are the dominant participants in all the dynamic energy consumption, while its usage decreases with the

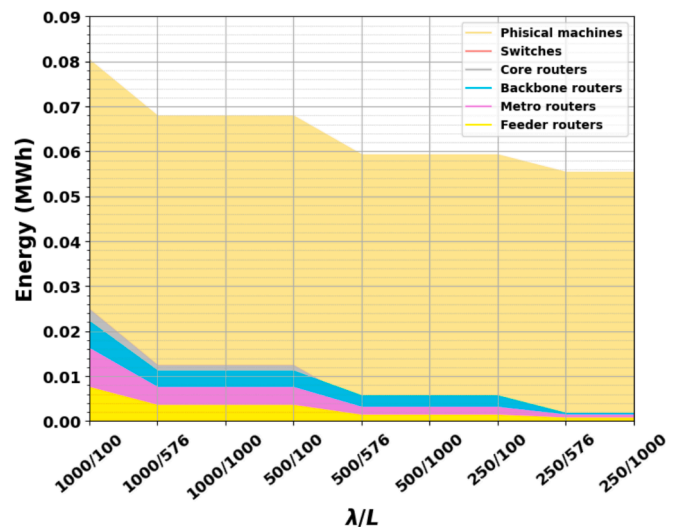


Fig. 11. Dynamic energy consumption of the system in a completely dispersed structure.

decrease of the λ/L ratio, which expresses efficiency while the system is spreading out. Switches participation- red area-it is possible to see that there is a drop in its energy usage while spreading is being increased. The cyan-for backbone routers, magenta for metro routers, and green for feeder routers-contributes much less to the overall energy consumptions compared with the physical machines and switches. Their energy consumptions go down when the λ/L is small. Overall, from this graph, when the system becomes more dispersive, the dynamic energy consumptions go down for all the components; hence, dispersion brings in energy efficiency.

4. Discussion

The telecommunication network that connects data centers and users represents a significant portion of overall energy consumption. This finding is consistent with existing research, which shows that telecommunication networks have a significant share in the overall energy consumption of ICT, including end-user devices [41]. Future advancements in distributed cloud architectures promise to minimize reliance on network routers by promoting local data processing wherever possible. Results indicate the clear impact of packet length on dynamic energy consumption in all architectures, especially during high traffic loads. Optimizing packet sizing strategies can significantly contribute to improving energy efficiency. Energy consumption efficiency plays a critical role in influencing the energy consumption of architectures that utilize medium to large data centers. Even advancements in ICT energy efficiency can be negated by high electricity consumption. Cloud service providers should prioritize strategies to minimize the effectiveness of electricity consumption in their data centers. In fully centralized and semi-distributed architectures, data center switches can account for more than half of the total energy consumed by the data center and up to 24 % of the total infrastructure energy consumption. Choosing the right network architecture and switch size is crucial for increasing energy efficiency in cloud infrastructure.

The fully distributed architecture eliminates the need for switches and data centers within the data center, leading to theoretically efficient power consumption. These characteristic positions fully distributed architectures and fully distributed architectures with central controllers as more energy-efficient compared to others. However, our analysis assumes a fixed performance of computational nodes in all architectures, which may not fully translate to future computational scenarios where heterogeneous end-user devices may be common. These findings emphasize the multi-faceted nature of optimizing cloud architecture for energy efficiency. Key considerations include:

- **Optimizing Telecommunication Network:** Strategies for minimizing energy consumption in the network infrastructure that connects data centers and users.
- **Packet length management:** Techniques for optimizing packet size for efficient data transmission.
- **Efforts to reduce the efficiency of electricity consumption:** Implementing measures to reduce the efficiency of electricity consumption in data centers, maximizing energy efficiency resulting from advances in device performance.
- **Designing data center networks:** Choosing network architecture and switch size to minimize energy consumption in data centers.

Also, real-world applications can be discussed. Most of them are reflected in the possible optimization of energy consumption in different cloud architectures, ranging from the fully centralized system to its decentralized counterpart. Since the energy model developed in this work considered both the static and dynamic use of energy, it would have a tool that would enable the cloud service providers to manage resources better. It will be used, for example, inside cloud data centers that will not use energy for cooling systems; it will balance workloads efficiently across servers, which reduces energy consumption on

network devices. Truly decentralized architectures will be deployable in highly dispersed computation settings, such as edge computing and fog computing, provided by applications of IoT-where the biggest energy consumption reduction will take place, from 19 % up to 28 % less when compared with a centralized system. These resulting power-saving habits mean greener, and thus cost-efficient, cloud services that will help service providers handle the increasing demands for data processing and storage with minimal environmental degradation.

5. Conclusion

As the landscape of cloud-related architectures continues to diversify, from fully centralized infrastructures to completely decentralized options, there is growing interest in determining which architecture offers superior energy efficiency. To address this pressing concern, this paper aims to shed light on the issue. It proposes a new classification system to categorize cloud-related architectures and highlight their initial topological distinctions. Building on this classification, a comprehensive, user-friendly, and scalable energy model is introduced, designed to accurately evaluate the energy consumption of different architectural configurations. This model precisely outlines the architectural differences in terms of energy usage and provides valuable insights into their relative efficiency. The study will conduct a critical review of the energy consumption pattern for a fully centralized, semi-distributed, fully distributed, and fully distributed with central controllers' cloud-related architecture. It is manifestly manifested through the energy profile given that a greater proportion is made up of static energy consumption in physical machines, switches, and routers. Fully centralized topologies were dominant, whereas the rest of the systems were quite similar; the load ratio based on flat consumption was about 1.3–1.4 MWh, whereas, for semi-distributed and fully distributed cases, physical machines deployed consumed up to 1.0–1.2 MWh static energy. However, dynamic energy consumption, much lower than static, depends on many dependencies such as system architecture and traffic load. For example, for fully distributed systems with central controllers, dynamic energy consumptions of the physical machines were less than 0.1 MWh, whereas in switches and routers, it is below 0.05 MWh. Physical machines are thus identified by the results as the main contributors to both the static and dynamic energy consumptions, especially for highly distributed configurations. This paper provides some empirical results: fully decentralized architectures attained up to 19–28 % energy consumption reduction compared to the fully centralized and semi decentralized architectures. In general, the work provided useful insight into the energy efficiency of alternative cloud architectures and provided ways to optimally shape cloud infrastructures with consideration for the minimum environment and operational cost. This energy model and classification framework may serve as a reference for further research and future decisions in designing energy-efficient cloud systems. In the future, cloud architecture development will be oriented toward optimization in saving energy, both for decentralized and semi-distributed systems, and with high-level integration of edge-fog computing for IoT applications. It also includes energy-aware routing and dynamic infrastructure adjustment in energy-efficient network design, while more efficient energy management techniques will further improve efficiency. In the near future, real-time energy optimization, enabled by AI and machine learning, will be possible. Green energy visibility and efficient hardware will characterize the sustainable data center. Hybrid architecture allows for flexible workload distribution at low energy consumption: centralized, distributed, and edge elements. Besides, new regulatory standards are going to guide and teach the ways of developing energy-efficient cloud infrastructures. Innovation will keep finding ways to handle continuous data demand with the highest energy efficiency possible.

This study offers valuable insights into the energy consumption of various cloud architectures, but it has certain limitations. It mainly focuses on energy metrics without considering the impact of varying

traffic patterns and network latency, which also affect energy efficiency. Additionally, the performance of decentralized systems in large-scale, real-world cloud environments remains untested, requiring further research on their scalability and robustness.

Future research could explore hybrid cloud architectures combining centralized and decentralized components to optimize energy efficiency. Investigating the integration of renewable energy sources, such as solar or wind, could further reduce environmental impact. Additionally, machine learning algorithms for dynamic energy optimization based on system loads and traffic, as well as the influence of emerging technologies like quantum and edge computing, offer promising directions for future studies.

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CRediT authorship contribution statement

Tangsen Huang: Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **Weiguo Huang:** Writing – review & editing, Writing – original draft, Software, Methodology. **Bin Zhang:** Writing – review & editing, Writing – original draft, Validation, Investigation. **Wenzhi Chen:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Xuewen Pan:** Writing – review & editing, Writing – original draft, Validation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] A. Shehabi, S. Smith, D. Sartor, R. Brown, M. Herrlin, J. Koomey, et al. United states data center energy usage report 2016.
- [2] R. Mahmud, R. Kotagiri, R. Buyya, Fog computing: a taxonomy, survey and future directions, *Internet of Everything: Algorithms, Methodologies, Technologies and Perspectives* (2018) 103–130.
- [3] C. Mouradian, D. Naboulsi, S. Yangui, R.H. Glitho, M.J. Morrow, P.A. Polakos, A comprehensive survey on fog computing: state-of-the-art and research challenges, *IEEE Commun. Surv. Tutorials* 20 (2017) 416–464.
- [4] M. Patel, B. Naughton, C. Chan, N. Sprecher, S. Abeta, A. Neal, Mobile-edge computing introductory technical white paper, White Paper, Mobile-Edge Computing (MEC) Industry Initiative 29 (2014) 854–864.
- [5] T. Taleb, K. Samdanis, B. Mada, H. Flinck, S. Dutta, D. Sabella, On multi-access edge computing: a survey of the emerging 5G network edge cloud architecture and orchestration, *IEEE Commun. Surv. Tutorials* 19 (2017) 1657–1681.
- [6] Y. Mao, C. You, J. Zhang, K. Huang, K.B. Letaief, A survey on mobile edge computing: the communication perspective, *IEEE Commun. Surv. Tutorials* 19 (2017) 2322–2358.
- [7] W. Shi, J. Cao, Q. Zhang, Y. Li, L. Xu, Edge computing: vision and challenges, *IEEE Internet Things J* 3 (2016) 637–646.
- [8] F. Bonomi, R. Milito, J. Zhu, S. Addepalli, Fog computing and its role in the internet of things, in: *Proceedings of the First Edition of the MCC Workshop on Mobile Cloud Computing*, 2012, pp. 13–16.
- [9] P. Hu, S. Dhelim, H. Ning, T. Qiu, Survey on fog computing: architecture, key technologies, applications and open issues, *J. Netw. Comput. Appl.* 98 (2017) 27–42.
- [10] D. Sabella, A. Reznik, R. Frazao, *Multi-access edge computing in action*, CRC Press, 2019.
- [11] R.N. Calheiros, R. Ranjan, A. Beloglazov, C.A.F. De Rose, R. Buyya, CloudSim: a toolkit for modeling and simulation of cloud computing environments and evaluation of resource provisioning algorithms, *Softw. Pract. Exp.* 41 (2011) 23–50.
- [12] G. Yao, D. Ding, C. Xie, H. Tan, Optimized design and comparative analysis of double-glazed photovoltaic windows for enhanced light harvesting and energy efficiency in cold regions of China, *Energy Build.* 325 (2024) 115011, <https://doi.org/10.1016/j.enbuild.2024.115011>.
- [13] H. Hosamo, G. Coelho, C. Nordahl Rolfes, D. Kraniotis, Building performance optimization through sensitivity analysis, and economic insights using AI, *Energy Build.* 325 (2024) 114999, <https://doi.org/10.1016/j.enbuild.2024.114999>.
- [14] S. Pan, C. Huang, J. Fan, Z. Shi, J. Tong, H. Wang, Optimizing internet of things fog computing: through lyapunov-based long short-term memory particle swarm optimization algorithm for energy consumption optimization, *Sensors* 24 (2024), <https://doi.org/10.3390/s24041165>.
- [15] Y. Shi, P. Chen, Energy retrofitting of hospital buildings considering climate change: an approach integrating automated machine learning with NSGA-III for multi-objective optimization, *Energy Build.* 319 (2024) 114571, <https://doi.org/10.1016/j.enbuild.2024.114571>.
- [16] M. As, T. Bilir, Machine learning algorithms for energy efficiency: Mitigating carbon dioxide emissions and optimizing costs in a hospital infrastructure, *Energy Build.* 318 (2024) 114494, <https://doi.org/10.1016/j.enbuild.2024.114494>.
- [17] A. Katal, T. Choudhury, S. Dahiya, Energy optimized container placement for cloud data centers: a meta-heuristic approach, *J. Supercomput.* 80 (2024) 98–140, <https://doi.org/10.1007/s11227-023-05462-2>.
- [18] C. Sharma, P.K. Tiwari, G. Agarwal, Energy-Efficient Resource Allocation Approaches for Cloud Computing Systems: A Survey and Taxonomy, in: A. K. Somani, A. Mundra, R. Doss, S. Bhattacharya (Eds.), *Smart Systems: Innovations in Computing*, Springer, Singapore, 2022, pp. 479–484.
- [19] J. Jang, J. Han, S.-B. Leigh, Prediction of heating energy consumption with operation pattern variables for non-residential buildings using LSTM networks, *Energy Build.* 255 (2022) 111647, <https://doi.org/10.1016/j.enbuild.2021.111647>.
- [20] A.S. Sabyasachi, J.K. Muppala, Cost-effective and energy-aware resource allocation in cloud data centers, *Electronics (basel)* 11 (2022), <https://doi.org/10.3390/electronics11213639>.
- [21] R.V. Klyuev, I.D. Morgoev, A.D. Morgoeva, O.A. Gavrina, N.V. Martyushev, E. Afremenkov, et al., Methods of forecasting electric energy consumption: a literature review, *Energies (Basel)* 15 (2022), <https://doi.org/10.3390/en15238919>.
- [22] A. Katal, S. Dahiya, T. Choudhury, Energy efficiency in cloud computing data centers: a survey on software technologies, *Cluster Comput.* 26 (2023) 1845–1875, <https://doi.org/10.1007/s10586-022-03713-0>.
- [23] R. Jeyaraj, A. Balasubramaniam, N. Guizani, A. Paul, Resource management in cloud and cloud-influenced technologies for internet of things applications, *ACM Comput. Surv.* 55 (2023), <https://doi.org/10.1145/3571729>.
- [24] B.T. Tabarsi, A. Rezaee, A. Movaghar, ROGI: partial computation offloading and resource allocation in the fog-based IoT network towards optimizing latency and power consumption, *Cluster Comput.* 26 (2023) 1767–1784, <https://doi.org/10.1007/s10586-022-03710-3>.
- [25] L. Mao, R. Chen, H. Cheng, W. Lin, B. Liu, J.Z. Wang, A resource scheduling method for cloud data centers based on thermal management, *Journal of Cloud Computing* 12 (2023) 84, <https://doi.org/10.1186/s13677-023-00462-2>.
- [26] W. Li, R. Tang, S. Wang, A fully distributed robust optimal control approach for air-conditioning systems considering uncertainties of communication link in IoT-enabled building automation systems, *Energy Build. Environ.* 5 (2024) 446–454, <https://doi.org/10.1016/j.enbenv.2023.02.001>.

- [27] A. Algarni, I. Shah, A.I. Jehangiri, M.A. Ala'Anzy, Z. Ahmad, Predictive energy management for docker containers in cloud computing: a time series analysis approach, *IEEE Access* 12 (2024) 52524–52538, <https://doi.org/10.1109/ACCESS.2024.3387436>.
- [28] R. Thaqi, M. Kadriu, B. Krasniqi, B. Rexha, On-premises versus cloud computing: a comparative analysis of energy consumption, *International Conference on Renewable Energies and Smart Technologies (REST) 2024* (2024) 1–5, <https://doi.org/10.1109/REST59987.2024.10645419>.
- [29] E.T. Martínez Beltrán, Á.L. Perales Gómez, C. Feng, P.M. Sánchez Sánchez, S. López Bernal, G. Bovet, et al., Fedstellar: a platform for decentralized federated learning, *Expert Syst. Appl.* 242 (2024) 122861, <https://doi.org/10.1016/j.eswa.2023.122861>.
- [30] A.H. Tariq, S.A.A. Kazmi, M. Hassan, S.A. Muhammed Ali, M. Anwar, Analysis of fuel cell integration with hybrid microgrid systems for clean energy: a comparative review, *Int. J. Hydrogen Energy* 52 (2024) 1005–1034, <https://doi.org/10.1016/j.ijhydene.2023.07.238>.
- [31] O. Ethan, H. Umar, Comparative Analysis of E-commerce Database Technologies: Blockchain, Scalable Storage, and Cyber Defense Strategies. Unpublished <https://doi.org/10.13140/RG.2024.2>.
- [32] D. Dhabliya, S.S. Dari, N.N. Sakhare, A.K. Dhabliya, D. Pandey, B. Muniandi, et al., New proposed policies and strategies for dynamic load balancing in cloud computing. emerging trends in cloud computing analytics, scalability, and service models, *IGI Global* (2024) 135–143.
- [33] F.C. Heinrich, T. Corneize, A. Degomme, A. Legrand, A. Carpen-Amarie, S. Hunold, et al. Predicting the energy-consumption of mpi applications at scale using only a single node. 2017 IEEE international conference on cluster computing (CLUSTER), IEEE; 2017, p. 92–102.
- [34] J. Logeshwaran, N. Shanmugasundaram, J. Lloret, Energy-efficient resource allocation model for device-to-device communication in 5G wireless personal area networks, *Int. J. Commun Syst* 36 (2023) e5524.
- [35] K. Liu, A. Zhao, J. Pan, Data Center Architecture, Operation, and Optimization. In: Toy M, editor. *Future Networks, Services and Management: Underlay and Overlay, Edge, Applications, Slicing, Cloud, Space, AI/ML, and Quantum Computing*, Cham: Springer International Publishing; 2021, p. 185–212. https://doi.org/10.1007/978-3-030-81961-3_6.
- [36] P.Y. Dibal, E.N. Onwuka, S. Zubair, E.I. Nwankwo, S.A. Okoh, B.A. Salihu, et al., Processor power and energy consumption estimation techniques in IoT applications: a review, *Internet Things* 21 (2023) 100655.
- [37] Y. Li, A.-C. Orgerie, J.-M. Menaud, Opportunistic scheduling in clouds partially powered by green energy, in: 2015 IEEE International Conference on Data Science and Data Intensive Systems, 2015, pp. 448–455.
- [38] F. Jalali, Energy consumption of cloud computing and fog computing applications, University of Melbourne, Department of Electrical and Electronic Engineering, 2015.
- [39] F. Etro, The economics of cloud computing. *Cloud technology: concepts, methodologies, tools, and applications*, IGI Global (2015) 2135–2148.
- [40] V. Sivaraman, A. Vishwanath, Z. Zhao, C. Russell, Profiling per-packet and per-byte energy consumption in the NetFPGA Gigabit router, in: 2011 IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), IEEE, 2011, pp. 331–336.
- [41] W. Van Heddeghem, S. Lambert, B. Lannoo, D. Colle, M. Pickavet, P. Demeester, Trends in worldwide ICT electricity consumption from 2007 to 2012, *Comput Commun* 50 (2014) 64–76.